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Project Title: GPS-Denied Navigation Assistance Module

1.Project Idea

This project focuses on developing a **GPS-Denied Navigation Assistance Module** that helps a operator reach a specified destination when GPS signals are unavailable, weak, or unreliable.

Unlike a conventional autonomous navigation system, this module does not directly control the controller. Instead, it uses sensor and localization data to estimate the position relative to the destination and provides simple movement suggestions to the operator through a display.

Based on the estimated position and movement, the system can provide instructions such as:

- Move Forward
- Move Left
- Move Right
- Move Backward
- Stop
- Destination Reached

A Raspberry Pi serves as the main processing unit of the system. It handles localization, navigation, decision-making, and the display of movement instructions.

Background

Autonomous platforms are widely used in applications such as search and rescue, surveillance, inspection, environmental monitoring, and disaster management. Most existing navigation systems rely heavily on GPS to determine the vehicle's position and guide it toward a destination.

However, GPS signals are not always reliable. In some environments, they may be weak, inaccurate, or completely unavailable. Examples include:

- Indoor environments
- Urban areas surrounded by tall buildings
- Tunnels or enclosed spaces
- Dense forests
- Disaster zones

- Areas affected by GPS interference or signal blockage

In these situations, it can be difficult for an operator to determine the correct direction in which the platform should move to reach its destination.

This project aims to address that problem by developing a navigation assistance module that can estimate the platform's relative movement and provide directional guidance without depending entirely on GPS.

Significance

The project offers a practical and modular approach to assisting navigation in environments where GPS cannot be relied upon.

Some of the key benefits of the system include:

- Supports navigation in GPS-denied or GPS-unreliable environments.
- Provides simple and easy-to-understand movement instructions to the operator.
- Reduces the difficulty of manually determining the correct direction toward a destination.
- Uses a Raspberry Pi as a cost-effective processing platform.
- Does not interfere with the vehicle's existing flight control system.
- Can be integrated with different platforms as an external navigation assistance module.
- Provides a foundation for future development of autonomous GPS-denied navigation systems.

2. Literature Survey

GPS is widely used for navigation and localization, but its signals may be unavailable or unreliable in environments such as indoors, tunnels, dense urban areas, and disaster zones. Therefore, alternative localization techniques are required for navigation in GPS-denied environments.

Visual Odometry and IMU Navigation

Visual Odometry (VO) estimates relative movement by analysing consecutive camera frames using techniques such as feature detection, feature tracking, and optical flow. However, visual methods can be affected by poor lighting, motion blur, and low-texture environments.

An IMU, consisting mainly of an accelerometer and gyroscope, provides acceleration and orientation information. However, errors accumulate over time, resulting in sensor drift and inaccurate position estimation.

Visual-Inertial Sensor Fusion

To overcome the limitations of individual sensors, researchers combine camera and IMU data using Visual-Inertial Odometry (VIO). Systems such as MSCKF, VINS-Mono, and HybVIO demonstrate that sensor fusion can improve the accuracy and robustness of relative motion estimation. However, advanced visual-inertial systems may require significant computational resources.

SLAM and Embedded Computing

SLAM techniques, such as ORB-SLAM3, can estimate position while building a map of the environment. Although effective for GPS-denied localization, full SLAM systems can be computationally complex for low-cost embedded devices. Therefore, a lightweight approach using efficient visual processing and IMU data is more suitable for a Raspberry Pi-based prototype.

Navigation Assistance

Most existing GPS-denied navigation systems focus on autonomous robots, drones, or vehicles. In contrast, navigation assistance systems provide understandable guidance to a human user instead of directly controlling the platform. Research on human-centred navigation shows the value of converting localization information into simple and actionable instructions.

8. Comparative Analysis of Existing Studies

Study	Year	Main Technique	Sensors Used	Main Contribution	Limitation	Relevance to Proposed System
Mourikis & Roumeliotis	2007	MSCKF	Camera + IMU	Efficient filtering-based visual-inertial estimation	Drift can remain without global correction	Supports lightweight sensor fusion
Scaramuzza & Fraundorfer	2011	Visual Odometry	Monocular/Stereo Camera	Established key concepts in visual motion estimation	Monocular scale ambiguity	Supports camera-based relative motion estimation
Qin et al.	2018	VINS-Mono	Monocular Camera + IMU	Robust visual-inertial state estimation	Computationally intensive	Demonstrates benefits of camera and IMU fusion
Campos et al.	2021	ORB-SLAM3	Camera + IMU	Visual-inertial localization and mapping	High computational requirements	Demonstrates advanced GPS-denied localization
Chen et al.	2021	Semantic Visual SLAM	RGB-D Camera + IMU	Navigation information for human assistance	More complex and costly hardware	Supports human-centred navigation guidance
Pulkkinen et al.	2022	HybVIO	Camera + IMU	Real-time visual-inertial tracking	Reduced complexity may involve accuracy trade-offs	Supports embedded VIO feasibility

Kühne et al.	2024	Optical Flow VIO	Camera + IMU	Reduced visual processing cost for embedded systems	Requires specialised sensing hardware	Highlights importance of lightweight visual processing
Mascetti et al.	2025	Navigation Guidance	Wearable/Smartphone Sensors	Discrete and actionable navigation instructions	Focused on pedestrian navigation	Supports movement guidance design

The comparative studies show a clear progression from visual motion estimation and inertial navigation toward visual-inertial fusion, SLAM, and human-centred navigation guidance.

9. Research Gap

The reviewed literature identifies three areas relevant to the proposed project.

9.1 Autonomous Navigation versus Navigation Assistance

A significant portion of GPS-denied navigation research focuses on fully autonomous systems for drones, robots, and vehicles. These systems connect localization outputs directly to path planners, controllers, and actuators.

In contrast, the proposed project focuses on **navigation assistance**. The system estimates relative movement and provides directional guidance to a user without controlling the platform. This creates a simpler, modular architecture that can potentially be adapted to different applications.

9.2 Advanced Localization versus Low-Cost Embedded Implementation

State-of-the-art systems such as VINS-Mono and ORB-SLAM3 provide sophisticated visual-inertial localization capabilities but can require considerable computational resources.

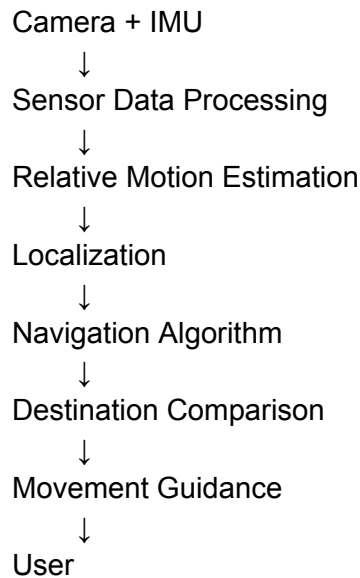
For a Raspberry Pi-based prototype, a lightweight implementation is more practical. The literature supports the use of efficient feature tracking and simplified visual-inertial estimation for resource-constrained systems.

9.3 Position Estimation versus Actionable Guidance

Many localization systems produce continuous outputs such as position coordinates, orientation, and pose estimates. However, raw numerical information is not always directly useful for a human user.

The proposed project explores the conversion of relative localization information into discrete and understandable movement instructions.

Therefore, the project investigates the following pipeline:



The project does not claim to replace advanced autonomous navigation or SLAM systems. Instead, it focuses on developing a **low-cost, general-purpose navigation assistance module** that translates sensor-based localization into useful movement guidance.

This positioning is directly consistent with the gaps identified in the reviewed survey.

10. Conclusion

The literature demonstrates that reliable navigation in GPS-denied environments cannot depend solely on a single sensing modality. Visual methods provide environmental information and relative motion estimation but can be affected by lighting, texture, and motion conditions. IMU-based navigation provides high-frequency motion information but suffers from accumulated drift. Combining these sensors through visual-inertial processing can improve the robustness of relative localization.

Advanced systems such as VINS-Mono and ORB-SLAM3 demonstrate the capabilities of visual-inertial state estimation and SLAM. However, their complexity may be unsuitable for a

short-duration, low-cost Raspberry Pi prototype. Lightweight visual-inertial techniques and efficient feature tracking provide a more practical direction for embedded implementation.

Based on the reviewed literature, the proposed system will focus on a practical architecture in which a **camera and IMU provide visual and inertial data to a Raspberry Pi**, which estimates relative movement, determines the direction of a predefined destination, and generates simple navigation guidance for the user.

3. Problem Statement & Objectives

Problem Statement

In GPS-denied or GPS-restricted environments, operators may face difficulty in determining the correct direction and movement required to reach a destination.

Existing autonomous navigation systems can be complex because they require direct integration with controllers, autonomous path planning, stabilization, and safety mechanisms.

There is a need for a lightweight and modular navigation assistance system that can estimate the relative direction and distance to a destination and provide simple movement guidance to the operator without directly controlling the platform.

Objectives

The objectives of the project are:

1. To design a GPS-denied navigation assistance module for autonomous navigation applications.
2. To estimate the relative position of the platform using onboard sensors and localization techniques.
3. To calculate the direction and distance between the current position and the target destination.
4. To process localization data using a Raspberry Pi.
5. To generate simple movement suggestions such as Forward, Left, Right, Backward, or Stop.
6. To display the suggested movement direction and navigation information to the operator.
7. To develop the system as a modular unit that does not directly control the platform.
8. To create an architecture that can be extended in the future using Visual Odometry, Visual-Inertial Odometry, or SLAM.

4. Methodology & System Architecture

4.1 Methodology

The proposed methodology consists of six major stages.

Step 1: Sensor Data Acquisition

The system collects data from the available onboard sensors.

The primary sensor configuration may include:

- Camera for visual information
- IMU for acceleration and orientation data

Depending on the prototype configuration, additional sensor data can be integrated later.

Step 2: Localization

The acquired sensor data is processed to estimate the relative position and orientation of the vehicle.

For the initial prototype, the system can implement a simplified localization approach. The architecture will be designed so that more advanced techniques such as Visual Odometry or Visual-Inertial Odometry can be integrated later.

The localization module provides information such as:

Current Position:

$X = x$

$Y = y$

$Z = z$

Orientation:

Heading = θ

Step 3: Destination Input

The destination is represented in a local coordinate system.

For example:

Current Position:
(0, 0)

Target Position:
(10, 5)

The system compares the current position with the target position.

Step 4: Navigation Computation

The Raspberry Pi calculates:

- Difference in X coordinate
- Difference in Y coordinate
- Distance to the target
- Relative direction to the target

The distance can be calculated using:

Distance Formula

The direction is determined based on the relative position of the target with respect to the current orientation.

Step 5: Movement Decision Logic

The system converts navigation information into simple movement instructions.

Example:

Target is ahead:
→ MOVE FORWARD

Target is to the left:
→ MOVE LEFT

Target is to the right:
→ MOVE RIGHT

Target is behind:
→ MOVE BACKWARD

Target reached:
→ STOP

The system may also display combinations or directional priorities based on the navigation algorithm.

Step 6: Display Output

The final movement suggestion is displayed to the operator.

The operator uses this information to manually control the platform.

4.2 System Architecture

